TemocTrade

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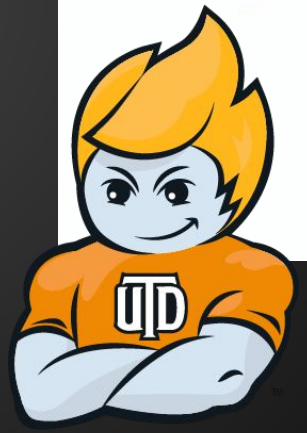
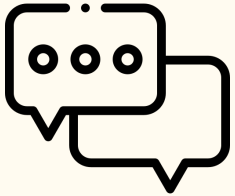


Objective



- The Marketplace for UT Dallas students, by UT Dallas students
- Ensures a safe environment
- Buy, sell, and bid on items
- Supports on-campus meetups or secure shipping
- Real-time auctions for in-demand listings
- Direct messaging between buyers and sellers
- A convenient, trustworthy, and student-focused marketplace

Cost Estimation

- Method of Estimate: Application Composition
- Hardware Costs: Our only hardware in this project that we already don't have, would be purchasing the domain for the application, in which for an annual cost, it would be around $10-20 annually.
- Personnel Costs: ~$7000

Software Cost Estimation

Cost Breakdown

- Development Tools → **Free**
- Frameworks → **Free**
- Database → **~$700/year**
- Cloud Hosting → **~$600/year**
- Design Tools → **~$144/year**

Additional Tools

- Microsoft Office → **~$70/year**
- Adobe Creative Cloud → **~$720/year**
- ChatGPT → **~$240/year**

Total Cost

$2,400 – $2,500 per year

Project Timeline

Requirement Analysis
1 week
2/9/26 - 2/13/26

Design
1 week
2/16/26 - 2/27/26

Frontend & Backend Development
8 weeks
3/2/26 - 4/10/26

Quality & Testing
2 weeks
4/13/26 - 4/24/26

Deployment & Training
1 week
4/27/26 - 5/1/26

- 7 hours per week for 14 weeks
  - Starting February 9 until May 1 of 2026
- Weekends excluded

Functional requirements

The User must be able to:

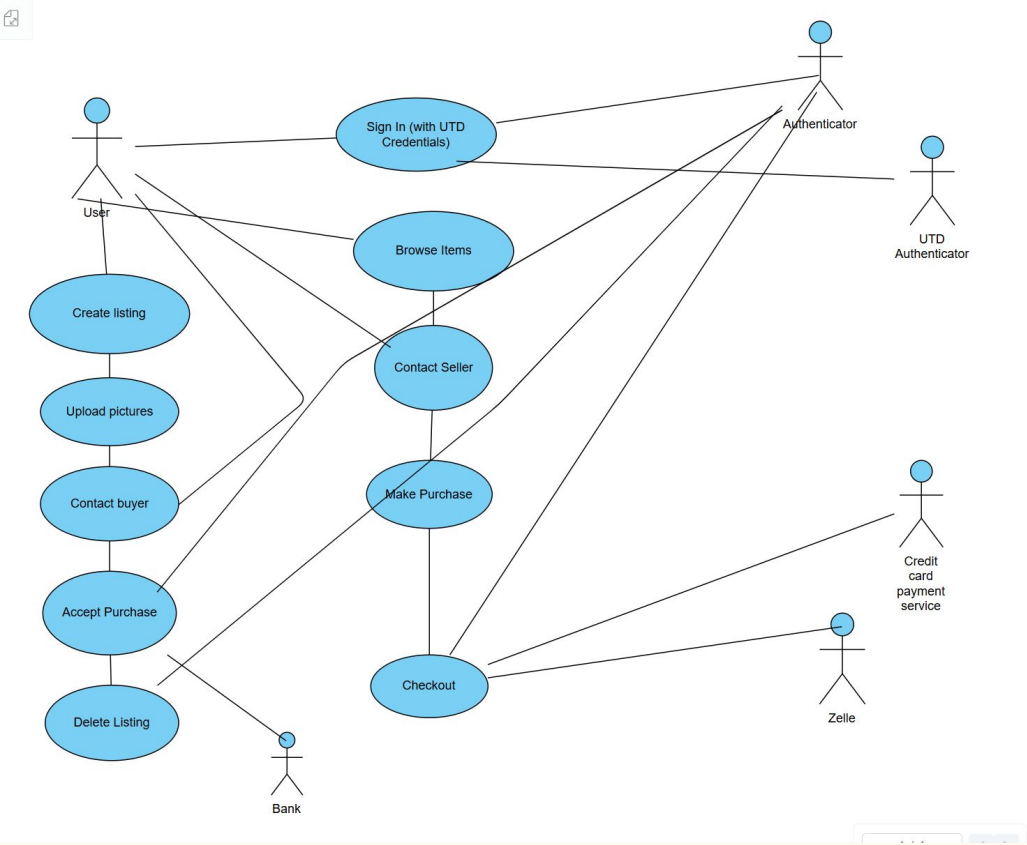
- authenticate using a UTD email
- create item listings
- manage their listings
- search and browse listings
- purchase items through payment gateway
- message sellers

Non Functional requirements

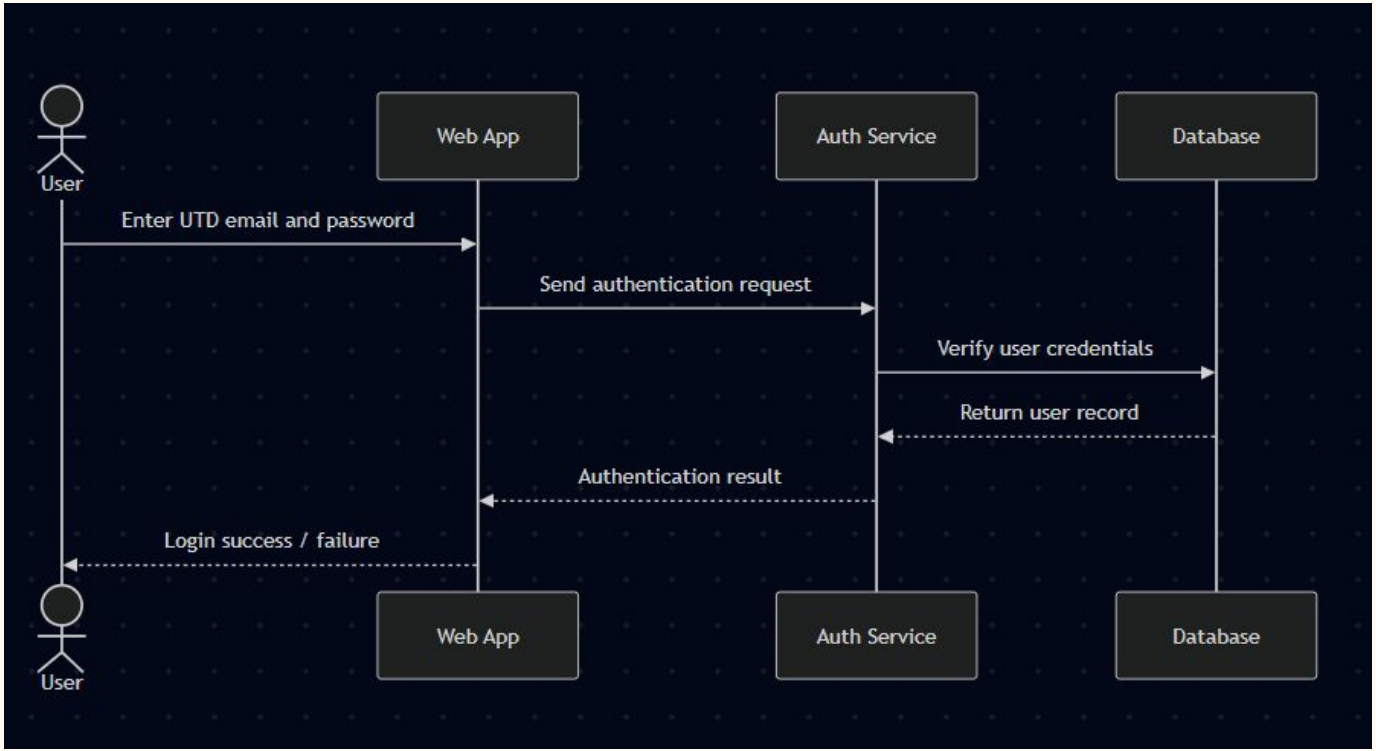
- Usability : User interface must be simple to navigate. Any feature should be reachable within 5 clicks
- Performance : User searches should return results in under 2 seconds with a reasonable amount of active users.
- Space: The application must not be over 200 MB.
- Dependendability: System should be available during 99% of usage times
- Security: Personal information and payments detailed will not be shown and encrypted.
- Environmental: The application shall be accessible through both mobile devices and on desktop.
- Operational: The system shall be open and functional at all times of the day, at all days of the week.
- Development: All code must be developed according to the pre-determined tech stack.
- Regulatory: PCI DSS. The system does not store payment information after a transaction.
- Ethical: The platform must not allow inappropriate behavior or content from users.
- Accounting: The system must safely log all transactions, bids, sales, and activity.
- Safety : Users will have a way to report other users who are scamming

Use case diagram

- Actors (User/authenticators/payment handlers)
- Use cases (signing in, creating a listing, browsing items, contacting seller/buyer, make purchase, etc.)
- Users can be both sellers and buyers
- Puts into perspective the interactions of the app, and the entities that do so with it.



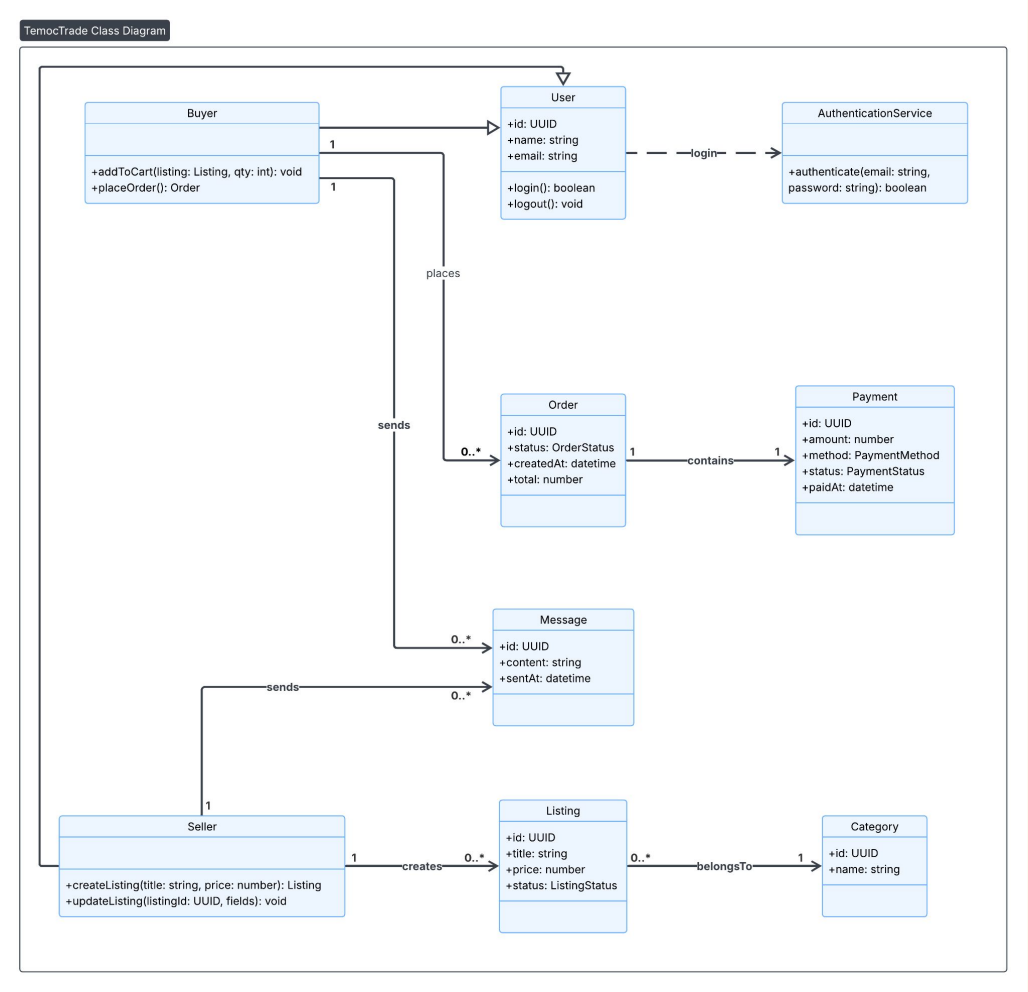
Sequence diagram



Class Diagram

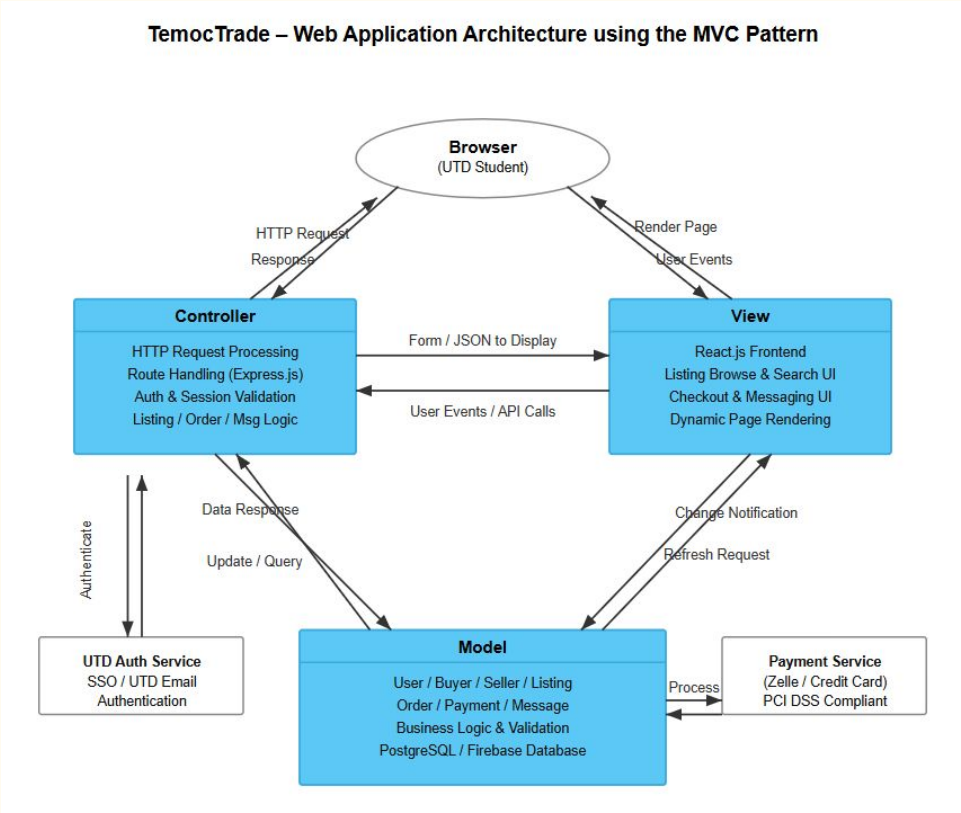
Key Components

- User (Buyer & Seller roles)
- Listings & Categories
- Orders & Payments
- Messaging System
- Authentication Service



Architectural Design

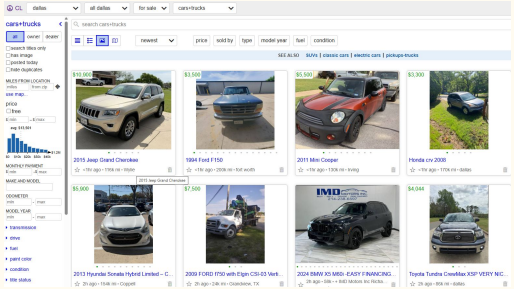
TemocTrade follows the Model-View-Controller (MVC) architecture pattern. The system is separated into three main components: Model, which handles business logic and database operations; View, which represents the UI; and the Controller, which processes user activity by interacting with the Model and updating the View. This architecture choice improves maintainability and scalability of the system.



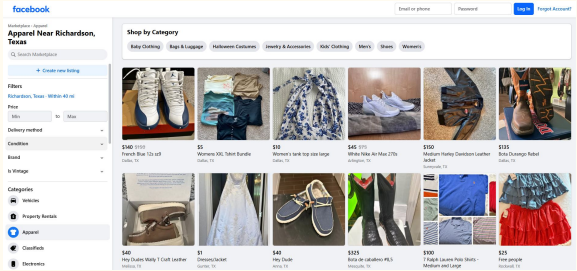
Design Comparisons

Comparison with Existing Platforms

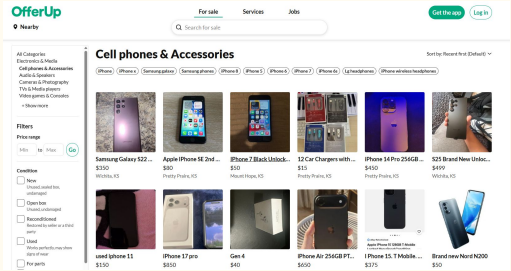
Craigslist



Facebook Marketplace



OfferUp



Design Comparisons

Platform Comparison

Craigslist

- Simple & outdated UI
- No strong authentication

Facebook Marketplace

- Large user base
- Messaging + profiles
- Can be cluttered

OfferUp

- Modern mobile design
- Local transactions
- Open to all users

Design Comparisons

Why TemocTrade?

Key Advantages:

- Verified users (UTD email)
- Safer transactions
- Focused student marketplace
- Less clutter, more relevant listings

Test Case execution

The unit we will test is the validate_listing_price. The function should return true for an int or float greater than 0 and false for anything that is not an int or float, or 0 or less. We shouldn't allow users to have negative pricing for products, or input strings or another data type. We will test this using the python library unittest.

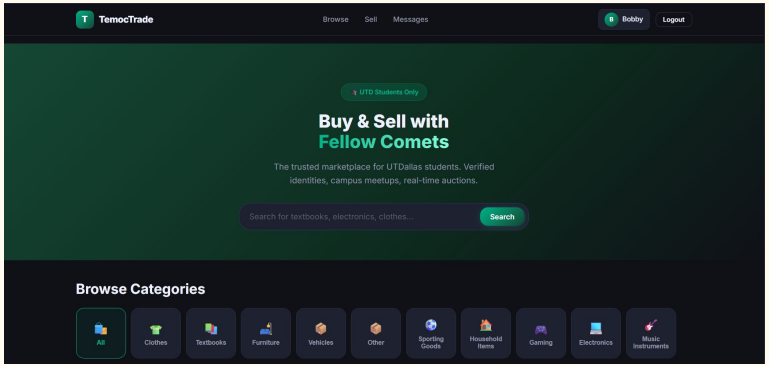
```
def validate_lisiting(price):
    if isinstance(price, (int, float))
        and price >0:
        return True
    else:
        return False
```

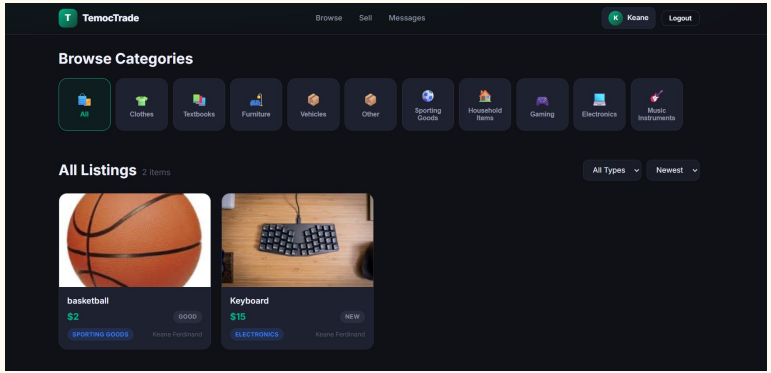
```
PS C:\Users\kdfer> & C:/Users/kdfer/AppData/Local/Programs/Python/Python313/python.exe c:/Users/kdfer/Downloads/test.py
.....
----------------------------------------------------------------------
Ran 5 tests in 0.001s

OK
PS C:\Users\kdfer>
```

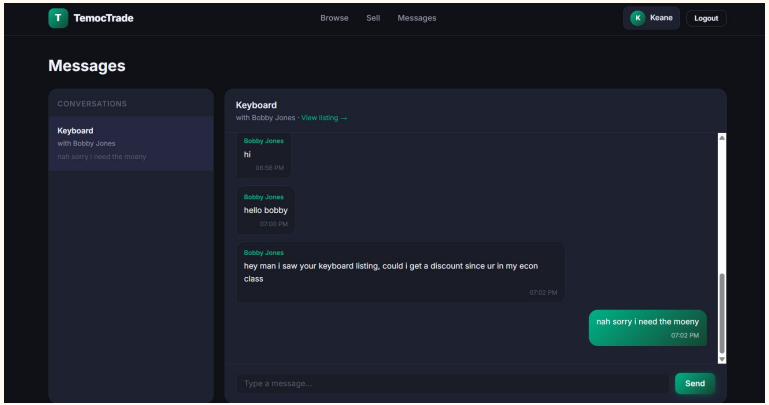
Test Case code

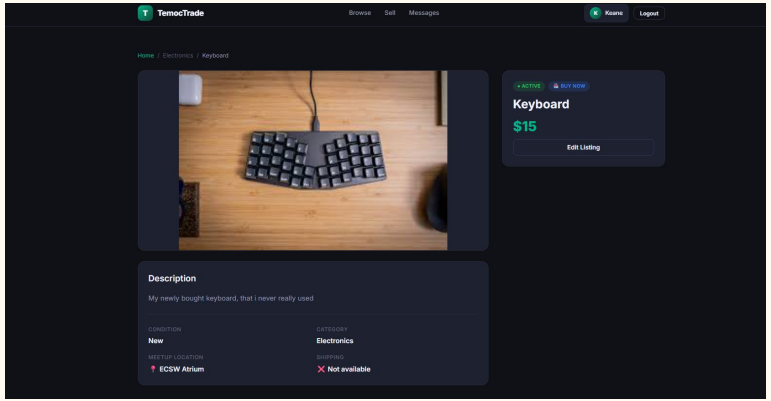
```
import __main__
import unittest # python library which allows for unit testing
from market_logic import validate_listing_price # bring in our module
to test
class TestMarketLogic (unittest.TestCase): # class to wrap different
test cases
    def test_valid_price(self): # test valid
        self.assertTrue(validate_listing_price(15.50))
    def test_negativeInvalid_price(self): # test negative
        self.assertFalse(validate_listing_price(-2.00))
    def test_zero_price(self): # test 0
        self.assertFalse(validate_listing_price(0))
    def test_string_input(self): # test string input
        self.assertFalse(validate_listing_price("twenty dollars"))
    def test_none_price(self): # test none
        self.assertFalse(validate_listing_price(None))
if __name__ == '__main__': # means you only run when script is run
directly
    unittest.main()
```

Prototype


TemocTrade
Browse
Sell
Messages
Bobby
Logout
UTD Students Only
Buy & Sell with
Fellow Comets
The trusted marketplace for UTDallas students. Verified identities, campus meetups, real-time auctions.
Search for textbooks, electronics, clothes...
Search
Browse Categories
All
Clothes
Textbooks
Furniture
Vehicles
Other
Sporting Goods
Household Items
Gaming
Electronics
Music Instruments


TemocTrade
Browse
Sell
Messages
Keane
Logout
Browse Categories
All
Clothes
Textbooks
Furniture
Vehicles
Other
Sporting Goods
Household Items
Gaming
Electronics
Music Instruments
All Listings 2 items
All Types
Newest
basketball
$2
GOOD
SPORTING GOODS
Keane Ferdinand
Keyboard
$15
NEW
ELECTRONICS
Keane Ferdinand


TemocTrade
Browse
Sell
Messages
Keane
Logout
Messages
CONVERSATIONS
Keyboard
with Bobby Jones
nah sorry i need the moeny
Keyboard
with Bobby Jones · View listing →
Bobby Jones
hi
06:56 PM
Bobby Jones
hello bobby
07:00 PM
Bobby Jones
hey man i saw your keyboard listing, could i get a discount since ur in my econ class
07:02 PM
nah sorry i need the moeny
07:02 PM
Type a message...
Send


TemocTrade
Browse
Sell
Messages
Keane
Logout
Home / Electronics / Keyboard
ACTIVE
BUY NOW
Keyboard
$15
Edit Listing
Description
My newly bought keyboard, that i never really used
CONDITION
New
CATEGORY
Electronics
MEETUP LOCATION
ECSW Atrium
SHIPPING
Not available

Challenges and Lessons Learned

- Converting our idea into formal software design documents
- Matching up requirements, drawings, and architectures
- Estimating realistic costs and schedules
- Collaborating effectively among team members
- Scope management in a constrained schedule
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- Good initial planning saves on workarounds
- Proper delegation encourages teamwork
- Every design document must be in agreement
- Effective estimation adds realism to the planning process

Conclusion

TemocTrade evolved from an initial concept into a more complete software design

Our design became more realistic, structured, and implementation-focused

Overall, the project now better reflects a practical UTD student marketplace system

Sources Used

[1] S. MacDonald, “How much does a domain name cost?,” Squarespace, https://www.squarespace.com/blog/how-much-does-domain-name-cost#:~:text=The%20average%20cost%20of%20a,Newer%20extensions%2C%20such%20as%20. (accessed Apr. 5, 2026).

[2] Python Software Foundation, “unittest — Unit testing framework,” *Python Documentation*. [Online]. Available: https://docs.python.org/3/library/unittest.html. [Accessed: Apr. 5, 2026].

[3] Keyhole Software, “Custom Software Development Cost: 2026 Pricing & Timeline Benchmarks,” *Keyhole Software*, Jan. 14, 2026. https://keyholesoftware.com/cost-custom-software-development/. Accessed Apr. 5, 2026.

Claude for the implementation code